Journal of Aerosol Science 105 (2017) 151–165

Contents lists available at ScienceDirect

Journal of Aerosol Science

journal homepage: www.elsevier.com/locate/jaerosci

Brownian coagulation of a bi-modal distribution of both spherical and fractal aerosols

J. Burgalat[a,*], P. Rannou[a,b]

[a] GSMA, UMR 7331, BP1039, Université de Reims Champagne-Ardenne, 51687 REIMS cedex, France
[b] LATMOS, UMR CNRS 8190, Université de Versailles St-Quentin, Verrières le buisson, France

ARTICLE INFO

Keywords:
Brownian coagulation
Moment method
Fractal particles

ABSTRACT

The use of Global Circulation Models (GCM) or mesoscale models are now widely used to understand planetary climatic systems. The increasing complexity of the models and the more and more detailed observations of the planetary bodies necessitate a corresponding increase in the complexity of the physical processes which are included in these models. In all the planetary atmospheres in the solar system, aerosols and clouds can be found and therefore microphysical processes must be included in climate models. The most accurate models are those where aerosol and cloud droplet size distributions are described with bins. This type of sophisticated and accurate models only needs a small quantity of a priori information to be used. However, they are demanding in computational resource. They can only be used in GCM at the cost of very long, and sometime prohibitive, time of computation. Alternatively, an other class of microphysical models, based on the description of the aerosol and cloud distributions with moments of distribution can be used. But, they need first to be developed and compared with a more detailed model and they need an *a priori* information about the size distributions.

In this article, we describe the development of the microphysical equations to treat the interaction of two populations of aerosols, with different geometrical structures, written with moment of distributions, interacting through Brownian coagulation. This problem was solved specifically for the case of Titan, the larger satellite of Saturn, where small aerosols have a spherical shape, large aerosols have a fractal aggregated structure and where aerosols bear an electric charge. The two populations interact in the mesosphere. The fractal structure of the large aerosols also has a consequence on the shape of the size distribution and on the laws of the microphysics that must be accounted *a priori* in the method. The case of Titan is probably one of the most complex and, in this work, we have written the set of equations in the most general way so they can be used for any other cases. Once developed, we finally compare the results yielded by our new model with the results obtained with the classical model based on a description in bins.

1. Introduction

Aerosols in planetary environments are frequent and can have a major impact on the climate. Beside the specific case of the Earth, it is also the case of several bodies: Mars, Titan, Venus, Pluto or giant planets. The increasing use of global climate models to study these planetary systems generally necessitate a complete description of processes related to aerosols (e.g. Kahre, Murphy, Haberle,

* Corresponding author.
E-mail address: jeremie.burgalat@univ-reims.fr (J. Burgalat).

http://dx.doi.org/10.1016/j.jaerosci.2016.11.009
Received 30 November 2015; Received in revised form 1 September 2016; Accepted 21 November 2016
Available online 26 November 2016
0021-8502/

Montmessin, & Schaeffer, 2005; Rannou, Hourdin, & McKay, 2002). The most straightforward technique to simulate aerosol microphysics is obtained with a model where the aerosol distributions are discretized with radius bins (e.g. Toon, Turco, Westphal, Malone, & Liu, 1988). In principle, this approach allows to describe all the microphysical laws without approximations, it allows to handle different types of interacting aerosol population and it needs a minimum *a priori* knowledge about the aerosols themselves. The required accuracy of the result can be controlled by a change in the discretization in the model. Therefore this type of model is suitable for planetary environment where the *a priori* knowledge about the aerosols is limited. The main drawback of this approach is the amount of computations needed to reach a result.

In the case of Titan, which is our main motivation for this work, models described with bins were used in 1D dimensional model (Cabane, Rannou, Chassefiere, & Israel, 1993; Toon, Turco et al., 1988), in the LMDZ-Titan 2D GCM (Rannou et al., 2002) and more recently in 3D GCM (Lebonnois, Burgalat, Rannou, & Charnay, 2012; Larson, Toon, & Friedson, 2014, 2015). In this planetary environment, the aerosols also serve as cloud condensation nuclei and then cloud microphysics must also be accounted for, increasing significantly the computational time (e.g. Barth & Toon, 2004; Rannou, Montmessin, Hourdin, & Lebonnois, 2006). This has become an obstacle for the further development of three dimensional climate model. An alternative to the bins is a description of the distributions with the moments of the function which describes the granulometry. In this case, a good *a priori* knowledge is needed about the size distribution and also about the aerosol properties. On Titan, previous sophisticated models allowed to constrain the aerosol distribution and structure. Titan aerosols have a unique specificity; they have a photochemical origin and they coagulate in such a way that large aerosols get a fractal aggregated structure (Cabane, Chassefiere, & Israel, 1992; Tomasko et al., 2008; West & Smith, 1991). As main consequences, the writing up of the microphysical laws must be adapted to account for this peculiar structure (Cabane et al., 1993). It modifies the mechanical properties of the aerosols and consequently the size distribution strongly departs from the classical log-normal distributions generally found in natural environments (Cours et al., 2011; Larson et al., 2014; Rannou, Hourdin, McKay, & Luz, 2004).

The scope of this work is to develop a microphysical model of aerosols with distributions defined by two moments, with size distributions of any form, and for spherical and fractal aggregate aerosols.

The model of Rannou et al. (2002) implemented in the LMDZ-Titan 2D GCM is used as reference model and we aim here to define the same processes: Our model primarily accounts for the Brownian coagulation and the sedimentation processes. Additionally, a production process is also defined. Each process is treated separately:

$$\frac{\mathrm{d}}{\mathrm{d}t}C(r, z) = Q_{production}(r, z) + \frac{\mathrm{d}}{\mathrm{d}t}C(r, z)\bigg|_{\text{sedimentation}} + \frac{\mathrm{d}}{\mathrm{d}t}C(r, z)\bigg|_{\text{coagulation}}$$

where $Q_{production}(r, z)$ is the production term, $C(r, z)$ is the aerosol concentration of radius r at the altitude z.

Although the development of this model is dedicated to the case of Titan, we developed the theoretical aspects in such a way that it can be used in other environments, as on the giant planets.

The paper is divided into four sections. After this Introduction, the second section is dedicated to the transformation of the microphysical laws in order to obtain a description with the moment of the distribution, rather than the distribution. We also describe how the secondary processes (sedimentation and production) are accounted for. In the third section, we present comparisons between the model described in moment and the reference bin model. We first show the performance of the model without neutral aerosols. Then, we show how the electric charges on aerosols affect the result with the two types of model. Finally we give a conclusion in the last section.

2. Theoretical aspects: Brownian coagulation

We present here the general coagulation equations for a bimodal size-distribution of different aerosols described with their moments. Moments are defined by

$$M_n(f) = \int_I x^n n(x)\mathrm{d}x \tag{1}$$

Each mode of the population represents particles with a specific structure (spherical or fractal) and a specific associated size-distribution law.

The aim of the present study is to define the set of general time-evolution equations for the distributions moments through Brownian coagulation process ranging from the continuous to the free-molecular flow regime. The development of the equations is performed in three main stages:

In a first step, the key of the method is to be able to write all the microphysics equations as a function of power laws of the particle radii and of the size of distributions. This necessitates, for some processes, some approximations that we describe in the following.

In a second step, we apply a simple substitution as described by Eq. (2) that allows to transform any integral where the particle radii only appear in power functions or in the size-distribution laws, as a product and sum of moments as follows:

$$\begin{aligned} I &= \int_0^\infty \int_0^\infty (r_i^p + r_j^k) n_S(r_i) n_F(r_j) \mathrm{d}r_i \mathrm{d}r_j = \int_0^\infty \int_0^\infty r_i^p n_S(r_i) n_F(r_j) \mathrm{d}r_i \mathrm{d}r_j + \int_0^\infty \int_0^\infty r_j^k n_S(r_i) n_F(r_j) \mathrm{d}r_i \mathrm{d}r_j \\ &= \int_0^\infty r_i^p n_S(r_i)\mathrm{d}r_i \int_0^\infty n_F(r_j)\mathrm{d}r_j + \int_0^\infty n_S(r_i)\mathrm{d}r_i \int_0^\infty r_j^k n_F(r_j)\mathrm{d}r_j I = M_p^S M_0^F + M_0^S M_k^F \end{aligned} \tag{2}$$

where n_S and n_F are two distributions of different particles (for instance, with S for spherical particles and F for fractal particles) and

Table 1
List of variables used in the equations.

Nomenclature	
$\alpha_S(x)$	Inter-moment relationship of the spherical mode
$\alpha_F(x)$	Inter-moment relationship of the fractal mode
η	Air viscosity
λ_g	Air mean free path
ρ_p	Aerosols bulk density
$A = 1.591$	Constant used in the slip flow correction approximation
b_k^i	b coefficient of the free molecular regime coagulation kernel
$C = A \times \lambda_g$	Slip flow correction factor
D_f	Fractal dimension of aerosols
$E = r_m^{\frac{D_f-3}{D_f}}$	Apparent to bulk radius conversion factor
k	Boltzmann constant
$K_{CO} = \dfrac{2k_BT}{3\eta}$	Thermodynamic pre-factor of the continuous regime
$K_{FM} = \left(\dfrac{6k_BT}{\rho_p}\right)^{1/2}$	Thermodynamic pre-factor of the free-molecular regime
r_a	Apparent radius of aerosol
r	Bulk radius of aerosol
$r_{C_S} = \left(\dfrac{1}{\alpha_S(3)}\dfrac{M_3^S}{M_0^S}\right)^{1/3}$	Spherical mode characteristic radius
$r_{C_F} = \left(\dfrac{1}{\alpha_F(3)}\dfrac{M_3^F}{M_0^F}\right)^{1/3}$	Fractal mode characteristic radius

r_i and r_j are the running parameters to describe the particle radius.

Finally, this kind of integral can yield moments of any order. However in a physical model, we only work with few selected moments. Then we need to relate any moment which may appear as a result of an integral to one of the selected moments. For this purpose, we use the inter-moment relationship function $\alpha(k, p_{\{i=1,N\}})$ as presented by Burgalat, Rannou, Cours, and Rivière (2014). This relation depends upon the aerosol size-distribution law through the parameters p_i that describe it. In our work, the size-distribution law has a fixed shape, and then the parameters p_i defined once for all, and therefore the inter-moment relationship function can be expressed as a function of the order k only:

$$\frac{M_k}{M_0} = r_0^k \times \alpha(k) \tag{3}$$

By construction $\alpha(3) = 1$, thus r_0 is a mean radius of the distribution such as $r_0 = (M_3/M_0)^{1/3}$.

This relation ensures that we can always write the time derivative of the selected moments, due to any physical processes, as a function of these moments only. The nomenclature of variables used in all the equations defined in the following sections is summarized in Table 1.

2.1. General coagulation equation

As described by Whitby and McMurry (1997), the time-evolution of the kth order moment of an aerosol population undergoing coagulation process can be defined as follows:

$$\frac{\mathrm{d}M_k}{\mathrm{d}t} = \underbrace{\frac{1}{2}\int_0^\infty\int_0^\infty \beta(r_i, r_j)(r_i^3 + r_j^3)^{k/3} n(r_i)n(r_j)\,\mathrm{d}r_i\,\mathrm{d}r_j}_{\text{Gain}} - \underbrace{\frac{1}{2}\int_0^\infty\int_0^\infty \beta(r_i, r_j)(r_i^k + r_j^k) n(r_i)n(r_j)\,\mathrm{d}r_i\,\mathrm{d}r_j}_{\text{Loss}} \tag{4}$$

To highlight the distribution in two modes, we write $n(r)$ as the sum of the distributions for the spherical particles, n_S, and for the fractal particles n_F. The moments of order k of these two modes are expressed further as ${M_k}^S$ and ${M_k}^F$. Substituting $n = n_S + n_F$ into Eq. (4) and sorting the terms reveals the different modes of interaction of the bimodal coagulation:

$$\frac{\mathrm{d}M_k}{\mathrm{d}t} = \underbrace{\frac{1}{2}\int_0^\infty\int_0^\infty \beta_{SS}(r_i, r_j)[(r_i^3 + r_j^3)^{k/3} - r_i^k - r_j^k]n_S(r_i)n_S(r_j)\mathrm{d}r_i\,\mathrm{d}r_j}_{\text{Intra-modal coagulation, spherical mode}} + \underbrace{\frac{1}{2}\int_0^\infty\int_0^\infty \beta_{FF}(r_i, r_j)[(r_i^3 + r_j^3)^{k/3} - r_i^k - r_j^k]n_F(r_i)n_F(r_j)\mathrm{d}r_i\,\mathrm{d}r_j}_{\text{Intra-modal coagulation, fractal mode}} + \underbrace{\int_0^\infty\int_0^\infty \beta_{SF}(r_i, r_j)[(r_i^3 + r_j^3)^{k/3} - r_i^k - r_j^k]n_S(r_i)n_F(r_j)\mathrm{d}r_i\,\mathrm{d}r_j}_{\text{Inter-modal coagulation}} \tag{5}$$

where β_{SS}, β_{SF}, β_{FF} are the coagulation kernels of each kind of interactions between particles. They differ between each other because the particle structure in each mode is different. Before going further in the development of Eq. (5), we need to define some rules that will govern the interactions of the coagulation process within each mode and between them.

1. The particle resulting from the coagulation of two particles of the *spherical* mode can be assigned either to the *spherical* or *fractal* mode depending on a probability law.
2. The particle resulting from the coagulation of two particles of the *fractal* mode remains in the same mode.
3. The particle resulting from the coagulation of particles from two different modes (inter-modal coagulation) is assigned to the *fractal* mode.

Following the rules described above and introducing, and p_k as the probability for spherical aerosols to remain in their mode during spherical intra-coagulation process, Eq. (5) can be split to obtain the time-evolution equations for our variables of interest that is the kth order moment of each mode.

$$\frac{\mathrm{d}M_k^S}{\mathrm{d}t} = \frac{p_k}{2}\int_0^\infty\int_0^\infty \beta_{SS}(r_i, r_j)(r_i^3 + r_j^3)^{k/3}n_S(r_i)n_S(r_j)\mathrm{d}r_i\,\mathrm{d}r_j - \frac{1}{2}\int_0^\infty\int_0^\infty \beta_{SS}(r_i, r_j)(r_i^k + r_j^k)n_S(r_i)n_S(r_j)\mathrm{d}r_i\,\mathrm{d}r_j - \int_0^\infty\int_0^\infty \beta_{SF}(r_i, r_j)(r_i^k)n_S(r_i)n_F(r_j)\mathrm{d}r_i\,\mathrm{d}r_j \tag{6}$$

$$\frac{dM_k^F}{dt} = \frac{(1-p_k)}{2}\int_0^\infty\int_0^\infty \beta_{SS}(r_i, r_j)(r_i^3 + r_j^3)^{k/3}n_S(r_i)n_S(r_j)\mathrm{d}r_i\,\mathrm{d}r_j + \frac{1}{2}\int_0^\infty\int_0^\infty \beta_{FF}(r_i, r_j)[(r_i^3 + r_j^3)^{k/3} - r_i^k - r_j^k]n_F(r_i)\,n_F(r_j)\,\mathrm{d}r_i\,\mathrm{d}r_j + \int_0^\infty\int_0^\infty \beta_{SF}(r_i, r_j)[(r_i^3 + r_j^3)^{k/3} - r_j^k]n_S(r_i)n_F(r_j)\mathrm{d}r_i\,\mathrm{d}r_j \tag{7}$$

In Eq. (6), the intra-modal coagulation process in the right hand side (RHS) has been split in order to highlight the first rule described above. The first term corresponds to the fraction of spherical part that remains in the mode, the second term is related to the loss of particles. The last term of the equation corresponds to the loss of particles through inter-modal coagulation.

From Eq. (7), the first term of the RHS is directly related to the rule (3) and corresponds to the gain of fractal particles through intra-modal coagulation of the spherical mode. The second term describes the intra-modal coagulation of the fractal mode and the last one, the interactions between the two modes as defined by the rule (2).

2.2. Coagulation kernels

As pointed by Cabane et al. (1993), coagulation kernels must be written differently for spherical particles and fractal aggregates. The major difference with the classical kernel coagulation expression is the definition of the particle radii involved in the equations which is different depending on if they come from the mass or from the surface of the particles, respectively noted r and r_a hereafter.

In our model, a fractal particle is defined as an aggregate of spherical particles of same size named monomer. For aggregates with a fractal dimension D_f, between 2 and 3 and composed of N spherical monomers of radius r_m, we can write the apparent radius $r_a = N^{1/D_f}r_m$. By extension, the aerosol bulk radius r which is the radius of the equivalent compact sphere is defined by $r = N^{1/3}r_m$. From these relations, we can derive the relationship between the bulk radius and the apparent radius:

$$r_a = r^{3/D_f} \times r_m^{(D_f-3)/D_f} \tag{8}$$

For spherical particles, the fractal dimension is fixed to 3 and thus, we simply have $r_a = r$. In the following development, we will always substitute Eq. (8) to r_a in order to define equations as a sum of bulk radius power laws. The scope of these transformations is to set the coagulation kernels as a function of power of r_i and r_j. This will ensure the possibility to integrate them following Eq. (2).

2.2.1. Near-continuum regime

The regime is defined by the case where the mean free path of the atmospheric molecules (λ_g) is much smaller than the particle apparent radius (r_a), noted with the superscript CO hereafter, the coagulation kernel is formulated as follows (Cabane et al., 1993; Fuchs, 1964):

$$\beta^{CO}(r_i, r_j) = 4\pi(r_{a_i} + r_{a_j}) \times (D_i + D_j) \tag{9}$$

where D_i is the diffusion coefficient of the particle i.

From the classical theory, the diffusion coefficient is expressed as:

$$D_i = \frac{k_B T}{6\pi\eta r_{a_i}} \times (1 + Kn[1.257 + 0.4\exp(-1.1\ Kn_i^{-1})])$$

where $Kn_i = \frac{\lambda_g}{r_{a_i}}$ is the Knudsen number of the particle i, η the air viscosity, k_B the Boltzmann constant and T the temperature.

Unfortunately, the exponential term defined in the slip-flow correction cannot be directly expressed as a sum of radius power law. Thus, we only use the first order approximation of the slip-flow correction (Park, Lee, Otto, & Fissan, 1999):

$$D_i \approx \frac{k_B T}{6\pi\eta r_{a_i}} \times (1 + A \times Kn_i)$$

Following Park et al. (1999), we use the value of A to 1.591 to get an accurate correction in the near-continuum regime ($Kn \in [0, 5]$).

Under this approximation, the coagulation kernel can be easily expressed as a simple sum of bulk radius power laws:

$$\beta_{SS}^{CO}(r_i, r_j) = K_{CO}\left[2 + \frac{r_i}{r_j} + \frac{r_j}{r_i} + C\left(\frac{1}{r_i} + \frac{1}{r_j} + \frac{r_j}{r_i^2} + \frac{r_i}{r_j^2}\right)\right] \tag{10}$$

$$\beta_{FF}^{CO}(r_i, r_j) = K_{CO}\left[2 + \left(\frac{r_i}{r_j}\right)^{3/D_f} + \left(\frac{r_j}{r_i}\right)^{3/D_f} + \frac{C}{E}\left(\frac{1}{r_i^{3/D_f}} + \frac{1}{r_j^{3/D_f}} + \frac{r_j^{3/D_f}}{r_i^{6/D}} + \frac{r_i^{3/D_f}}{r_j^{6/D}}\right)\right] \tag{11}$$

$$\beta_{SF}^{CO}(r_i, r_j) = K_{CO}\left[2 + \frac{r_i}{E\ r_j^{3/D_f}} + \frac{E r_j^{3/D_f}}{r_i} + C\left(\frac{1}{r_i} + \frac{1}{E r_j^{3/D_f}} + \frac{E r_j^{3/D_f}}{r_i^2} + \frac{r_i}{E^2 r_j^{6/D_f}}\right)\right] \tag{12}$$

where $E = r_m^{\frac{(D_f-3)}{D_f}}$ is a pre-factor introduced by the substitution of Eq. (8), $K_{CO} = 2k_B T/3\eta$, $C = A \times \lambda_g$ a term related to the slip-flow correction and D_f the fractal dimension of the fractal mode.

2.2.2. Free-molecular regime

The classical formula of the coagulation kernel in the free-molecular regime is written as follows (Friedlander, 2000):

$$\beta^{FM}(r_i, r_j) = \left(\frac{6k_B T}{\rho_p}\right)^{1/2} \times (r_{a_i} + r_{a_j})^2\sqrt{r_i^{-3} + r_j^{-3}} \tag{13}$$

In the free-molecular regime (noted with the superscript FM in the equations), a new difficulty arises. In Eq. (13), the term $\sqrt{r_i^{-3} + r_j^{-3}}$ cannot be expressed as a simple sum of radius power laws. As first suggested by Lee, Chen, and Gieseke (1984) and then used in other studies (Park et al., 1999; Pratsinis, 1988), we perform the following approximation:

$$\sqrt{r_i^{-3} + r_j^{-3}} \approx b \times (r_i^{-3/2} + r_j^{-3/2}) \tag{14}$$

where b is an effective factor to be computed numerically which depends on the interactions and on the order of the studied moments. In order to compute b, we must go back to Eqs. (6) and (7). The six integral terms can be expressed as follows:

$$I = \int_0^\infty \int_0^\infty \beta^{FM}(r_i, r_j) G(r_i, r_j, k) n(r_i)\, n'(r_j)\, \mathrm{d}r_i\, \mathrm{d}r_j$$

where $G(r_i, r_j, k)$ is a function of the particle radii found in the integrals (6) and (7), and also reported in Table 2. Using β^{FM} as expressed in (13) and introducing the approximation shown in Eq. (14), we find the expression of b:

Table 2
List of b_k coefficients and their associated $G(r_i, r_j, k)$ functions based on the different interactions between modes.

Id	$G(r_i, r_j, k)$ function	Associated equation (interaction)
$b_k^{(1)}$	$(r_i^3 + r_j^3)^{k/3}$	Eq. (6) – 1st integral (SS)
$b_k^{(2)}$	$r_i^k + r_j^k$	Eq. (7) – 1st integral (SS) Eq. (6) – 2nd integral (SS)
$b_k^{(3)}$	$(r_i^3 + r_j^3)^{k/3} - r_i^k - r_j^k$	Eq. (7) – 2nd integral (FF)
$b_k^{(4)}$	r_i^k	Eq. (6) – 3rd integral (SF)
$b_k^{(5)}$	$(r_i^3 + r_j^3)^{k/3} - r_j^k$	Eq. (7) – 3rd integral (SF)

$$b_k = \frac{\int_0^\infty \int_0^\infty \beta^{FM}(r_i, r_j) G(r_i, r_j, k) n(r_i) n'(r_j) \, \mathrm{d}r_i \, \mathrm{d}r_j}{\int_0^\infty \int_0^\infty \beta^{FM}(r_i, r_j) \frac{\left(r_i^{-\frac{3}{2}} + r_j^{-\frac{3}{2}}\right)}{\sqrt{r_i^{-3} + r_j^{-3}}} G(r_i, r_j, k) n(r_i) n'(r_j) \, \mathrm{d}r_i \, \mathrm{d}r_j} \tag{15}$$

The value of b_k depends on the integral of Eqs. (6) and (7) and on the value of k. Thus we have to define 10 values of b_k, for the five different G functions in integrals and the two orders of moments.

The consequence of such approximation implies that we now have to deal with five different kernels for the free-molecular regime. The coefficient also depends on the order studied. However, we can now expand the coagulation kernel formulae in radius power laws:

$$\beta_{SS}^{FM,(1)}(r_i, r_j, k) = b_k^{(1)} K_{FM} \left[r_i^{1/2} + 2\, r_i^{-1/2} r_j + r_i^{-3/2} r_j^2 + r_i^2 r_j^{-3/2} + 2\, r_i r_j^{-1/2} + r_j^{1/2} \right] \tag{16}$$

$$\beta_{SS}^{FM,(2)}(r_i, r_j, k) = b_k^{(2)} K_{FM} \left[r_i^{1/2} + 2\, r_i^{-1/2} r_j + r_i^{-3/2} r_j^2 + r_i^2 r_j^{-3/2} + 2\, r_i r_j^{-1/2} + r_j^{1/2} \right] \tag{17}$$

$$\beta_{FF}^{FM,(3)}(r_i, r_j, k) = b_k^{(3)} K_{FM} E^2 \left[r_i^{\frac{12-3D_f}{2D_f}} + 2\, r_i^{\frac{6-3D_f}{2D_f}} r_j^{3/D_f} + r_i^{-3/2} r_j^{6/D_f} + r_i^{6/D_f} r_j^{-3/2} + 2\, r_i^{3/D_f} r_j^{\frac{6-3D_f}{2D_f}} + r_j^{\frac{12-3D_f}{2D_f}} \right] \tag{18}$$

$$\beta_{SF}^{FM,(4)}(r_i, r_j, k) = b_k^{(4)} K_{FM} \left[r_i^{1/2} + 2\, E\, r_i^{-1/2} r_j^{3/D_f} + E^2\, r_i^{-3/2} r_j^{6/D_f} + r_i^2 r_j^{-3/2} + 2\, E\, r_i r_j^{\frac{6-3D_f}{2D_f}} + E^2\, r_j^{\frac{12-3D_f}{2D_f}} \right] \tag{19}$$

$$\beta_{SF}^{FM,(5)}(r_i, r_j, k) = b_k^{(5)} K_{FM} \left[r_i^{1/2} + 2\, E\, r_i^{-1/2} r_j^{3/D_f} + E^2\, r_i^{-3/2} r_j^{6/D_f} + r_i^2 r_j^{-3/2} + 2\, E\, r_i r_j^{\frac{6-3D_f}{2D_f}} + E^2\, r_j^{\frac{12-3D_f}{2D_f}} \right] \tag{20}$$

where $E = r_m^{\frac{(D_f-3)}{D_f}}$ is a pre-factor introduced by the substitution of Eq. (8), $K_{FM} = \sqrt{6k_BT/\rho_p}$ and D_f is the fractal dimension of the aerosols in the fractal mode. Here K_{FM} is a thermodynamic pre-factor with k_B the Boltzmann constant and ρ_p the aerosol bulk density (that is the density of the material composing the aerosols), that is the density of the material composing the aerosols.

Eqs. (16), (17) and (19) replace the coagulation kernel of each integral of Eq. (6) while Eqs. (16), (18) and (20) replace those of Eq. (7).

2.3. Time evolution of moments

Once the coagulation kernels in each flow regime have been expanded in power of r_i and r_j, we are able to establish the time-evolution equation for each moment of interest (order and mode) and for each flow regime.

To this point, no choice has been made on the order of the moments to study. Even if we could develop further the generic equations, we now write the final time-evolution equations for a two-moments scheme where the order 0 and 3 has been chosen. The reasons for this choice are the following:

- M_0 and M_3 represent convenient physical quantities of the aerosol population, respectively their total number and the total volume (to a factor $4/3\pi$).
- The development of the final set of equations is somehow simplified with these two moments, especially for the mass which is a conservative quantity.
- Using M_3 explicitly ensures the conservation of the mass during the coagulation process.

After a bit of calculation and substitution, one can easily show that, the time-evolution equations of the couple (M_0, M_3) for each mode can be reduced to a set of simple equations. The equations are casted under the same form whatever is the flow regime. Then, for a given flow regime, noted with a superscript X standing for CO or FM, we can write:

$$\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|^X = \underbrace{(p_0^X - 2)\gamma_0^{X,SS} \times M_0^{S^2}}_{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS}^X} + \underbrace{(-\gamma_0^{X,SF} \times M_0^S M_0^F)}_{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF}^X} \tag{21}$$

$$\left.\frac{\mathrm{d}M_3^S}{\mathrm{d}t}\right|^X = \underbrace{(p_3^X - 1)\gamma_3^{X,SS} \times M_3^{S^2}}_{\left.\frac{\mathrm{d}M_3^S}{\mathrm{d}t}\right|_{SS}^X} + \underbrace{(-\gamma_3^{X,SF} \times M_3^S M_3^F)}_{\left.\frac{\mathrm{d}M_3^S}{\mathrm{d}t}\right|_{SF}^X} \tag{22}$$

$$\left.\frac{\mathrm{d}M_0^F}{\mathrm{d}t}\right|^X = \underbrace{(1-p_0^X)\gamma_0^{X,SS} \times M_0^{S^2}}_{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS}^X} + \underbrace{(-\gamma_0^{X,FF} \times M_0^{F^2})}_{\left.\frac{\mathrm{d}M_0^F}{\mathrm{d}t}\right|_{FF}^X} \tag{23}$$

$$\left.\frac{\mathrm{d}M_3^F}{\mathrm{d}t}\right|^X = \underbrace{(1-p_3^X)\gamma_3^{X,SS} \times M_3^{S^2}}_{\left.\frac{\mathrm{d}M_3^S}{\mathrm{d}t}\right|_{SS}^X} + \underbrace{\gamma_3^{X,SF} \times M_3^S M_3^F}_{\left.\frac{\mathrm{d}M_3^F}{\mathrm{d}t}\right|_{SF}^X} \tag{24}$$

where ${p_k}^X$ is the probability for the spherical particles to remain in their mode after the intra-modal spherical coagulation (as introduced by the first coagulation rule) for the kth order moment. γ_k are terms that gather the pre-factors due to the expansion of the coagulation kernels (10)–(12), (16)–(20) and the conversion rules between moments displayed with Eq. (3).

2.4. Harmonic mean

Up to now, each tendency is given for a pure flow regime (continuous or free-molecular). In order to cover the entire range of particle size, including the transition regime, we applied the harmonic mean method used by Park et al. (1999) to the case of a bimodal size distribution. In our case, we compute the harmonic mean of the tendencies produced by each interactions (*SS*, *SF* and *FF*) defined in Eqs. (21)–(24) for each order and mode. As an example, for M_0^S the total number of aerosols of the spherical mode, from Eq. (21), we have:

$$\frac{\mathrm{d}M_0^S}{\mathrm{d}t} = \left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS} + \left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF}$$

With

$$\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS} = \frac{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS}^{CO} \times \left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS}^{FM}}{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS}^{CO} + \left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SS}^{FM}} = \frac{(p_0^{CO}-2)\gamma_0^{CO,SS} \times (p_0^{FM}-2)\gamma_0^{FM,SS}}{(p_0^{CO}-2)\gamma_0^{CO,SS} + (p_0^{FM}-2)\gamma_0^{FM,SS}} \times M_0^{S^2}\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF} = \frac{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF}^{CO} \times \left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF}^{FM}}{\left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF}^{CO} + \left.\frac{\mathrm{d}M_0^S}{\mathrm{d}t}\right|_{SF}^{FM}}$$
$$= -\frac{\gamma_0^{CO,SF} \times \gamma_0^{FM,SF}}{\gamma_0^{CO,SF} + \gamma_0^{FM,SF}} \times M_0^S M_0^F \tag{25}$$

The same rule is applied to the other case, giving then a complete set of tendencies for the moments of each modes.

2.5. Transfer probabilities

As already mentioned in the first coagulation rule, particles from the spherical mode are allowed to be transferred into the fractal mode through the spherical intra-modal coagulation. It describes the fact that two particles of the spherical mode can produce fractal aggregates.

A simple way to represent this aspect is to apply a probability (ranging from 0 to 1) that defines the proportion of coagulation that yields particles in each mode. The definition of this probability is quite straightforward: if the size of the resulting particle of the spherical intra-modal coagulation is less than or equal to the size of the monomer (which is, as a reminder, the size of the spherical grains building the aggregate), it remains in the spherical mode, otherwise, it is transferred to the fractal mode. The probability that the aerosols resulting from the coagulation SS remains in the mode S is defined as:

$$p_k = \frac{\int_0^{r_m} r^k\, F_s(r)\mathrm{d}r}{\int_0^{+\infty} r^k\, F_s(r)\mathrm{d}r} \tag{26}$$

where

$$F_s(r) = \frac{1}{2}\int_0^r \beta_{SS}(u,\, r-u)n(u)n(r-u)\mathrm{d}u$$

3. Additional processes

In order to develop a one-dimensional numerical model based on the theory presented in the previous sections, some other processes related to microphysics have to be updated to be suitable for a use in a moments-scheme such as the sedimentation and the production processes. Moreover, as our main goal is to define a model suitable for Titan atmosphere, we need to account for the electric charging effect of the aerosols (Borucki et al., 1987). Then, we also include in our model a correction used to account for the effect of the electric charging of aerosols onto the coagulation process. The following paragraphs describe the way we have handled these processes.

3.1. Electric charging of aerosols

When aerosols are electrically charged, the coagulation coefficient $\beta(r_i, r_j)$ should be adjusted by a multiplicative factor to account for electrical interactions of the particles involved in the coagulation process. In this work, we only deal with the effect of electrostatic repulsion, and the correction factor is defined as follows (Pruppacher & Klett, 1978). We use the same approximation as used by Cabane et al. (1993) and assume that r_a is the relevant radius to use in the correction:

$$Q(r_{a_i}, r_{a_j}) = \frac{y}{\exp(y) - 1}; \quad y = \frac{(n_{e^-} q_{e^-})^2 r_{a_i} r_{a_j}}{4\pi\varepsilon_0 (r_{a_i} + r_{a_j}) kT} \tag{27}$$

where n_{e^-} is the number of electrons per length unit, q_{e^-} the charge of an electron, ε_0 is the electric constant, k the Boltzmann constant and T the temperature.

Due to the exponential term, the electric charging correction cannot be expanded in terms of radius power law. No obvious simplification can be used here. As a workaround, we propose here to use the mean effect of the electrical charging correction for each coagulation sub-kernel described in the equations.

For a given interaction noted XY where X and Y refer to SS, SF or FF, we note the effective electrical charging correction of this coagulation process:

$$\overline{Q}_{XY} = \frac{\int_0^\infty \int_0^\infty Q(r_i, r_j)\, \beta_{XY}(r_i, r_j)\, n_X(r_i) n_Y(r_j)\, dr_i\, dr_j}{\int_0^\infty \int_0^\infty \beta_{XY}(r_i, r_j)\, n_X(r_i) n_Y(r_j)\, dr_i\, dr_j} \tag{28}$$

As $\overline{Q}_{XY}$ is weighted by the coagulation kernel, it is in fact a function of the pressure, temperature and the characteristic radius of the size-distribution involved in the interaction. For this calculation, the exact coagulation kernel can be used and the correction is applied directly to the harmonic tendencies of Eq. (21)–(24).

3.2. Sedimentation

Although we also have to deal with some approximations, the development of the time-evolution equation for the sedimentation process is far much simpler than for the coagulation equations. Indeed, it is expressed as:

$$\frac{\partial}{\partial t} M_k(z) = -\vec{\nabla} . \overrightarrow{\Phi_{sed}} = -\frac{d}{dz}\left(\int_0^{+\infty} n(r, z) \times \omega(r, z) r^k dr\right)$$

For a fractal dimension between 2 and 3, and using the first order approximation of the slip-flow correction, the settling velocity of a particle can be given as:

$$\omega(r, z) = \frac{2\rho_p g}{9\eta E} \times \left(r^{\frac{3D_f - 3}{D_f}} + \frac{A\lambda_g}{E} r^{\frac{3D_f - 6}{D_f}}\right) \tag{29}$$

where $E = r_m^{\frac{(D_f-3)}{D_f}}$ is a pre-factor introduced by the substitution of Eq. (8), A the coefficient used in the first order slip-flow correction factor and λ_g the atmospheric mean free path. The sedimentation flux of the kth order moment is obtained by applying Eq. (1):

$$\Phi_{M_k}(z) = \int_0^\infty r^k\, n(r, z)\omega(r, z)\, dr \Phi_{M_k}(z) = \frac{2\rho_p g}{9\eta E}\left(M_{\frac{D_f(k+3)-3}{D_f}} + \frac{A\lambda_g}{E} M_{\frac{D_f(k+3)-6}{D_f}}\right) \tag{30}$$

Sedimentation process is implemented in the model using a flux method as presented by Toon, McKay, Courtin, and Ackerman (1988). In order to limit numerical diffusion in the sedimentation process due to the vertical discretization, a correction is applied to the settling velocity as described in Fiadeiro and Veronis (1977). This correction is proved to be efficient in sectional models. However in our model, the correction is applied separately for the two moments (M_0, M_3), and it does impact the fluxes in a different way for the two moments. This breaks the consistency of the sedimentation for a given distribution, and it then produces strong bias. This is already a known problem (Meyers, Walko, Harrington, & Cotton, 1997) and to avoid it, we then choose to force the sedimentation speed of the two moments to be the same, this of the moment M_0.

3.3. Aerosol production

Our model is first intended to be used in climate modelling of Titan atmosphere. The injection of aerosols is performed at the top of model that corresponds to the production of photochemical aerosols at high altitude. The production process is represented in the model by a simple production law using a Gaussian function (Cabane et al., 1992; Toon, Turco et al., 1988):

$$\frac{dM_3^S}{dt} = \frac{Q_0}{\rho_p}\exp\left[-\frac{1}{2}\left(\frac{z-z_0}{\Delta z}\right)^2\right] \tag{31}$$

where Q_0 is the aerosol production rate in $\mathrm{kg\,m^{-3}\,s^{-1}}$, ρ_p the aerosol bulk density, z_0 the production altitude and Δz the standard deviation of the production zone.

In order to obtain the tendency of M_0^S, we simply use Eq. (3) with a r_C value that represents the radius of the aerosols produced by photochemistry, set as a free parameter of the model ($r_C = 2 \times 10^{-8}$ m for this work).

$$\frac{dM_0^S}{dt} = \frac{1}{r_C^3 \times \alpha(3)}\frac{dM_3^S}{dt}$$

4. Validation and tests

4.1. Models

To validate our model, we performed a comparison with a reference model described with bins (Cabane et al., 1992; Toon, Turco, et al., 1988). To finish the description of the model, we need to specify the size distributions that we use in the model with moments. The inter-moments relation (Eq. (3)) and the size-distribution laws have been computed in the same way as in Burgalat et al. (2014). We used the same inter-moment relation for both the spherical and fractal mode. This implicitly means that their size distribution laws are the same.

For practical convenience, the transfer probabilities, the b coefficient and the electric charging correction have been pre-calculated offline. In our case study, as we use the same size-distribution law for each mode with a fixed shape, each $b^{(k)}$ is a constant and then all of them are set as fixed parameters in the model. Transfer probabilities and electric charging correction are function of the temperature, the pressure, the charging rate (n_{e-} from Eq. (27)) and the characteristic radius of the mode involved in the relevant process. In the current version of the model, we have created look-up tables covering the whole range of possible values for these parameters. In our model, we do not allow dynamic variations of n_{e-} which remains fixed spatially and temporally for each simulation.

The production rate used for all simulations, as well as its associated characteristic radius, is fixed to $3.5 \times 10^{13}\ \mathrm{kg\,m^{-3}\,s^{-1}}$ and 2×10^{-8} m respectively.

The vertical structure in the model is defined as in Rannou et al. (2004). The altitudes are discretized following the levels of the GCM, with slabs of about 10 km in the middle stratosphere, decreasing to 5 km in the low stratosphere and few km in the troposphere.

Hereafter we denote the bin reference model as *model B* and the moments model as *model M*. For our tests, the size-distribution of the *model B* is casted in 100 bins in order to obtain the most accurate results as possible. In order to compare the two models, we need to define the range of spherical aerosols and fractal aerosols in the *model B*. We assume that the spherical mode covers the range from the first bin up to the bin corresponding to monomer radius. All other bins are part of the fractal mode. M_3 is simply computed as the sum of each bins concentration times the cube of the bin radius size.

4.2. Results and comparisons

We ran the two models (*B*, *M*) until they reached a convergence state which is characterized by the fact that for each vertical level, the results do not present time variations any more. Both models share the same input atmospheric profiles (temperature and pressure levels) taken from Burgalat et al. (2014). The atmospheric profile remains fixed during the simulation and no coupling between microphysics and radiative transfer is made. The moments for the aerosol distributions are set to zero at the beginning of the run.

The time simulation to reach the convergence state may vary depending on the value of the electric charging of aerosols used. The simulation time increases with the electric charging. For our tests, the simulation time varied from 350 (no electric charging) to 5900 terrestrial years ($n_{e^-} = 45\mathrm{e}^-\ \mu\mathrm{m}^{-1}$).

In the following part, we focus on the comparisons between the two models using the couple of moments (M_0 and M_3) for each mode and their sum which has a physical meaning since it represents the total population in terms of number (M_0) or volume (M_3).

Fig. 1 shows the vertical profiles of the moments for each mode and their sum. From the atmospheric profile used, the free-molecular regime corresponds to part of the atmosphere with pressure levels below $P \approx 1$ mbar. At pressure above about 1 mbar, the near-continuum and continuum regimes prevail.

In the free-molecular regime, the maximum of the vertical distribution of moments (M_0 and M_3) is always at higher altitude for the *model M* than the *model B* (Fig. 1a and c). Discrepancies for the fractal mode at high altitude ($P \leq 0.03$ mbar, Fig. 1c and d) denote the fact that the transfer probability laws are not enough representative: they act as starter processes that generate fractal particles from intra-coagulation of the spherical mode (first member of Eqs. (23) and (24)). These discrepancies directly impact the mean radius of each mode in the production zone, where the mode of spherical particles feeds the mode of fractal aerosols (Fig. 2).

Yet, some work on the spherical mode size-distribution has to be done in order to get more accurate results at high altitudes where both modes interact actively. However this bias does not prevent the *model M* to work in good agreement with the *model B* below the production zone. At lower altitudes ($P \geq 1$ mbar), the discrepancies between the two models are lower than 20% with an average

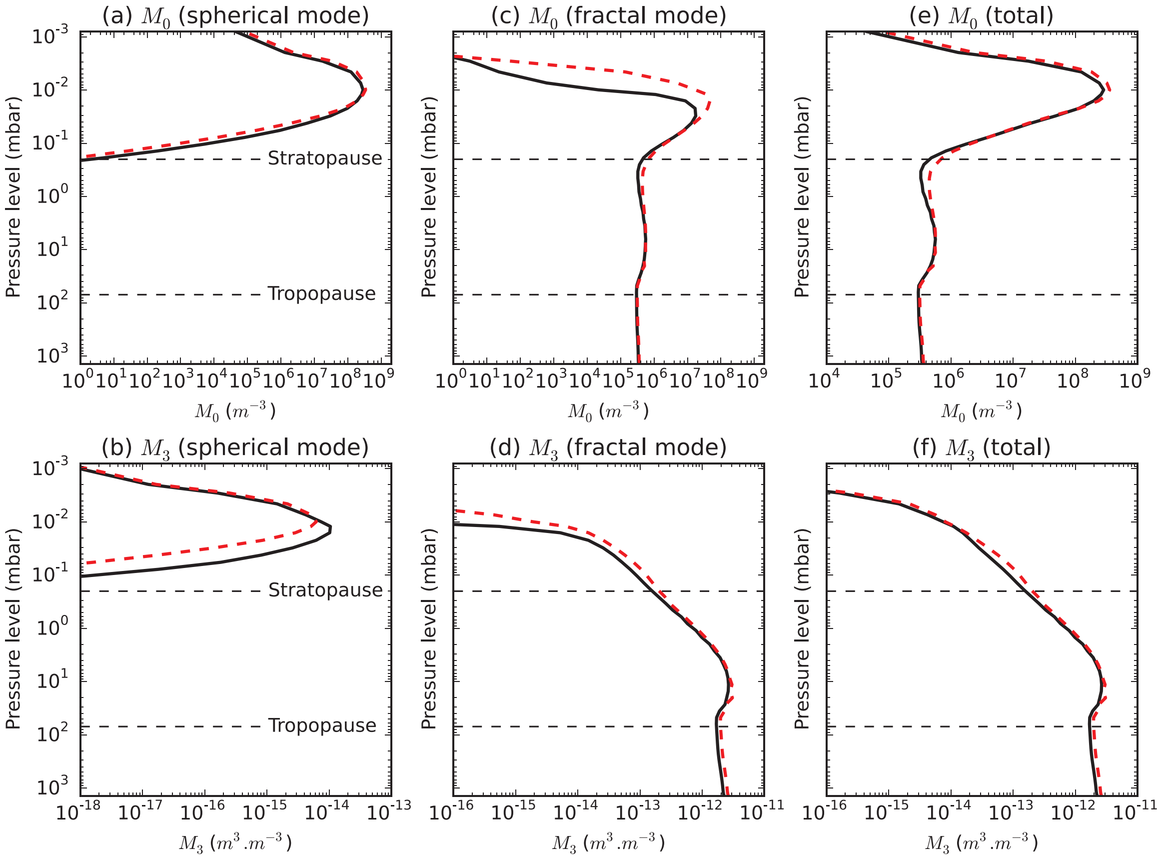

Fig. 1. Vertical profile of M_0 (top) and M_3 (bottom) for the reference sectional model (black, solid) and the moments model (red, dashed). Left figures (a,b) show the spherical mode (i.e. sum of all bins up to the monomer radius in the reference model). Center figures (c,d) show the fractal mode (remaining bins of the grid compared to the spherical mode). Right figures (e,f) represent the sum of the two modes. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

difference of 5% for M_0^F and 17% for M_3^F for the lower layers at pressure levels $P \geq 35$ mbar. At these altitudes, only the intra-coagulation of the fractal mode is activated because there are no more small aerosols. As discussed further, the approximation made on the slip-flow correction and the sedimentation process explains this relative error in the model, which is acceptable for a climate model.

Since our goal is to use the model for Titan global climate modelling, we also performed tests on the electrical charging correction

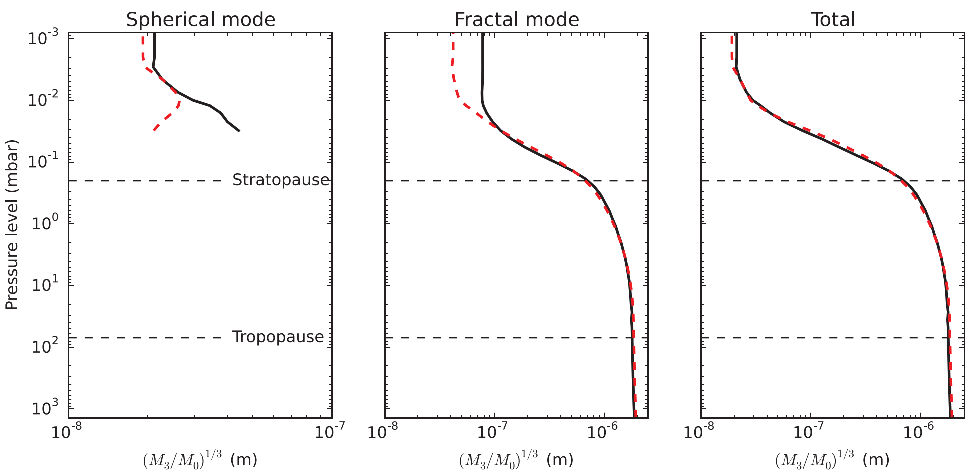

Fig. 2. Vertical profiles of the approximative mean radius ($\bar{r} = (M_3/M_0)^{1/3}$) for the reference sectional model (black, solid) and the moments model (red, dashed). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

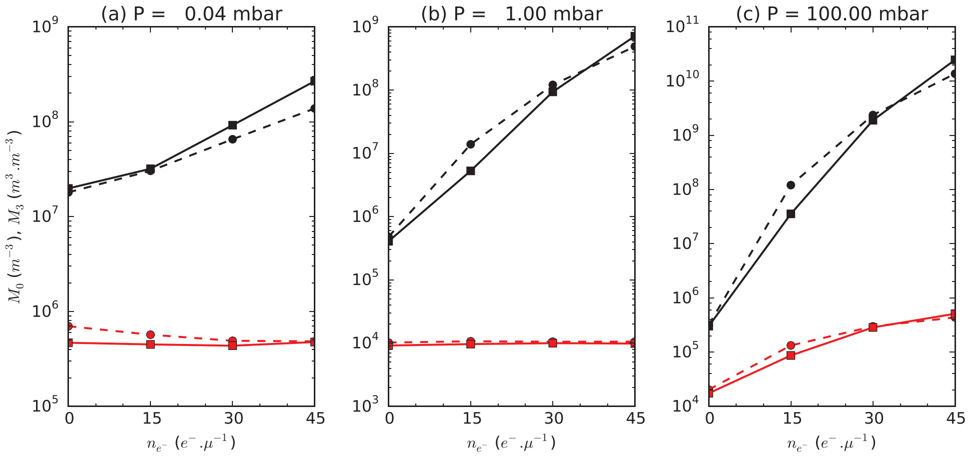

Fig. 3. Evolution of M_0 (black) and M_3 (red) for the reference model (solid line) and the moments model (dashed line) at three different pressure levels. M_3 curves are scaled by a multiplicative factor. M_3 is multiplied by 10^{19} for $P = 0.04$ mbar and by 10^{16} for $P = 1$ mbar and $P = 100$ mbar. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

as presented in the theoretical aspects (Eq. (28)). We ran four simulations with different values of the electric charge of aerosols ($n_{e^-} = 0,\ 15,\ 30,\ 45$ e$^-$ μm^{-1}). Fig. 3 shows the evolution of the total moments (M_0 and M_3) of each model as a function of the electric charge for different pressure levels. The models behave qualitatively in the same way: the total number of particles increases with the electric charge, which is the expected behaviour because electric charging tends to reduce the coagulation process. In the free-molecular regime, the total volume of aerosols presents no sensitivity in the middle stratosphere where there are only fractal aerosols (Fig. 3b) while we can observe a small sensitivity in the production zone, where spherical and fractal particles can be found (Fig. 3a). That is because the settling velocity of fractal aerosols in the free-molecular regime is not size discriminant and the coagulation process conserves the total volume of aerosols. On the other hand, sedimentation of the spherical aerosols is size-discriminant, and thus the amount of aerosols will depend on the charging effect.

At lower altitude (in the near-continuum regime), the settling velocity of fractal particles is also size-discriminant. We then do more not expect in this region a constant M_3^F with the charging rate. As the mean radius of the fractal mode decreases with the increasing electric charge, fractal aerosols settle slower and thus M_3^F increases with the electric charge as shown in Fig. 3c. Although *model M* has an overall correct behaviour, it shows some discrepancies in M_0 which goes from 10% for the best cases, up to a factor 3 for the worst case ($n_{e^-} = 15$ e$^-$ μm^{-1}). This performance should be discussed in regard with the fact that the total number of aerosols vary by several orders of magnitude under a change of the charging rate. As for previous works, the charging rate can be considered as a free parameter essentially because the real effect on the aerosol in Titan atmosphere is not well known and the electrical correction is generally applied uniformly on Titan while the real charging rate is indeed extremely variable with time and space (Borucki, Whitten, Bakes, Barth, & Tripathi, 2006). We find that a moderate offset of 5 e$^-$ μm^{-1} is enough to correct the difference between the two models if needed. As the electric charging is a free parameter which is not very well constrained on Titan, we consider that these discrepancies are acceptable considering the simple approximation of the mean effect made in the *model M* and may be easily compensated.

4.3. Discussion

We evaluate that the larger source of error in our model is due to the approximations performed in the sedimentation process to develop the description with moments. We first used an approximation of the slip-flow correction, using $1 + A \times Kn$ instead of $1 + Kn \times [1.257 + 0.4 \times \exp(-1.1Kn^{-1})]$, in order to lead the analytical calculation. Secondly, we used the same sedimentation speed for the two moments M_0 and M_3 in order to avoid instabilities, while their respective sedimentation speed, ω_{M_0} and ω_{M_3}, should

Table 3
Mean relative difference, $\frac{\Delta M_k}{M_k}$, of the total moment of the size distribution in different cases in the stratosphere (100 mbar–1 mbar).

Tested approx.[a,b]	1 vs 2 Slip-flow correction	2 vs 3 Settling velocity	2 vs 4 Settling velocity	1 vs 3 Global difference *model B* vs *model M*
$\frac{\Delta M_0}{M_0}$	+0.05	+0.05	+0.11	+0.09
$\frac{\Delta M_3}{M_3}$	−0.001	+0.17	+0.03	+0.17

[a] **"i vs j"** indicates that the model #j is compared to the model #i. The relative is given as $(M_j - M_i)/M_i$.
[b] List of models: **Model 1:** Reference bin model (*model B*). **Model 2:** Reference bin model with slip-flow correction ($1 + AKn$). **Model 3:** Moment model using the settling velocity ω_{M_0} relevant to M_0. (*model M*). **Model 4:** Moment model using the settling velocity ω_{M_3} relevant to M_3.

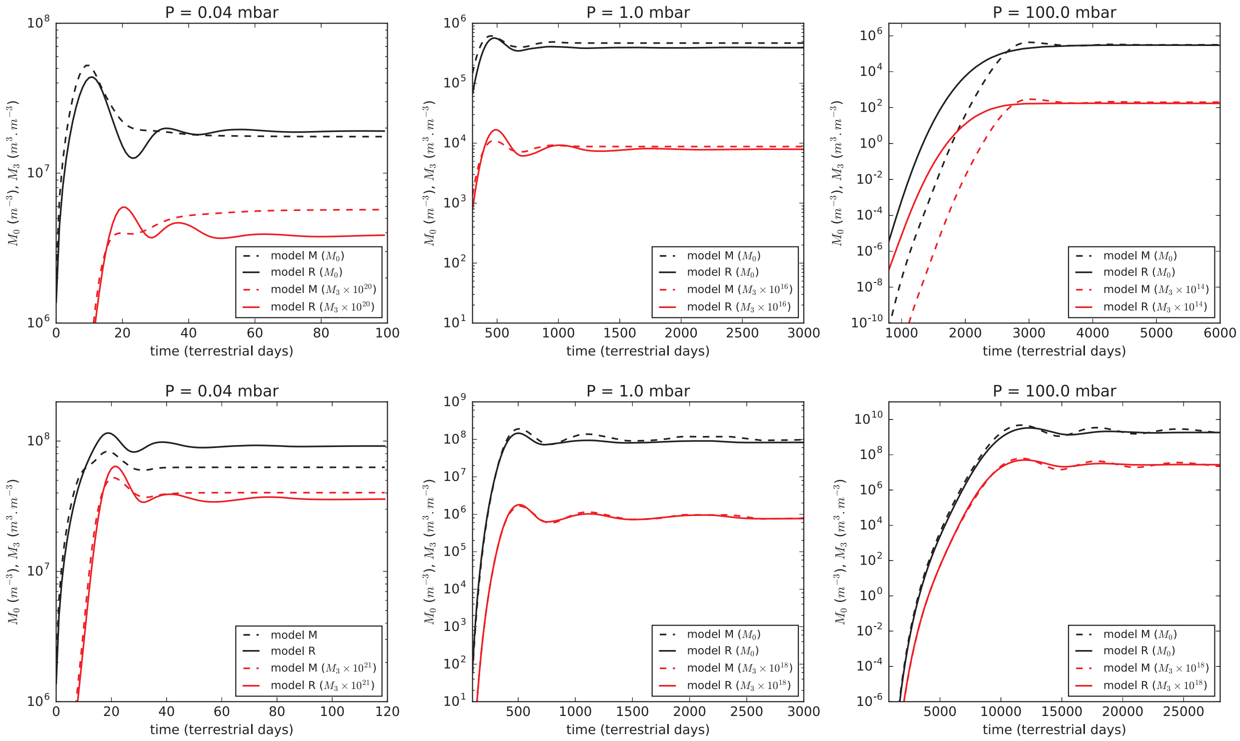

Fig. 4. Evolution of M_0 (black curves) and M_3 (red curves) for the reference sectional model (solid) and the moment model (dashed), for two values of the electric charging correction: $n_{e^-} = 0$ (top) and $n_{e^-} = 30\ \mathrm{e^-\ \mu m^{-1}}$ (bottom), during the transient period. All M_3 curves are scaled by a multiplicative factor given in the legend of each plot. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

indeed differ.

We evaluated the impact of the two approximations made in the sedimentation model (Table 3). The simplified expression of the slip-flow correction can be tested directly with the bin model. We find a difference of $\simeq +5\%$ and $\simeq -0.1\%$ on the moments M_0 or M_3 respectively (column 1 of Table 3). This clearly represents a small error for such a complex model. The use of the same settling speed can be evaluated by comparing the bin model with the simplified slip-flow correction and the moment model, with the same simplified slip-flow correction, and with either the sedimentation speed relevant to the moment M_0 or M_3 (ω_{M_0} and ω_{M_3} respectively). The results show that choosing a sedimentation velocity relative to a given moment allows to better conserve this moment than the other. In our model, we choose to used the sedimentation speed ω_{M_0} and the error on M_0 is 5% while the error on M_3 is 17% (column 2 of Table 3). The column 3 of the same table shows the performance we would have by choosing the speed ω_{M_3}.

We find that the sum of the errors due to the approximations (columns 1 and 2) are comparable to the global error between the *model B* and *model M* (last column of Table 3). Therefore, we conclude that the approximations we made in the sedimentation process explain much of the difference between the two models.

4.4. Evolution with time

So far, we only discussed the converged solutions of the models. Fig. 4 shows evolutions with time of the total moment (that is spherical + fractal) M_0 and M_3 for each model, for two values of n_{e^-} (0 and 30 $\mathrm{e^-\ \mu m^{-1}}$) during the transient period and for different pressure levels. Here the models start from an atmosphere free of aerosols. The results show that the evolution can differ between the models depending on the cases. In the *model M*, the choice to use the settling speed ω_{M_0} for both moments makes the aerosols to settle slower than it should. However, it is noteworthy that $\omega_{M_0} = \omega_{M_3}$ is valid in the molecular regime ($Kn \gg 1$) for fractal aerosols with a fractal dimension 2 because the sedimentation speed does not depend on the aerosol size. That is no more the case in the continuous regime ($Kn \ll 1$) where the sedimentation speed depends on the aerosol radius. Therefore, for this reason, the time evolution is more reliable at high altitude (low pressure levels) for all cases than at low altitude. At low pressure, the evolution of the moments for both models are similar even if the *model B* generates oscillations before reaching the steady state solution that may differ in the *model M*. At low altitude (level 100 mbar), there is a delay in populating the lower layers in the case $n_{e^-} = 0\ \mathrm{e^-\ \mu m^{-1}}$, but not in the case $n_{e^-} = 30\ \mathrm{e^-\ \mu m^{-1}}$. This is because increasing charging rates corresponds to decreasing aerosol radii. Thus, without charge, the aerosols reach the continuous regime at pressure below 100 mbar where the hypothesis $\omega_{M_0} = \omega_{M_3}$ breaks down. For the case with $n_{e^-} = 30\ \mathrm{e^-\ \mu m^{-1}}$, aerosols get a smaller radius due to coagulation inhibition and then reach the transition regime at lower altitude. The time evolution remains reliable down to and probably below 100 mbar (around the tropopause). We note here that the case with $n_{e^-} = 30\ \mathrm{e^-\ \mu m^{-1}}$ is more representative of the Titan case than the case with $n_{e^-} = 0\ \mathrm{e^-\ \mu m^{-1}}$. The difference in the details of the time evolution must be regarded having in mind that this is inherent to the method generally employed for moments (especially the choice of one settling speed for all moments). It should also be kept in mind that in the frame of a GCM, we always consider an atmosphere already loaded in aerosols and submitted to other types of transport (advection, diffusion).

5. Conclusion

We have presented, in this work, the development of equations which allow to deal with the Brownian coagulation of two modes of aerosols with different structures (spherical and fractal) and size distributions, described with two of their moments. Our main motivation is to produce an aerosol microphysical model described with moments that could replace a more demanding model described with bins in a Titan General Circulation Model – TGCM (Lebonnois et al., 2012; Rannou et al., 2004). In the present work, the description is adapted to the case of Titan haze, although the full description is general enough to be applied to the case of other planets.

For practical purpose, we have selected the moment of order 0 and 3 which are directly related to the number and volume of aerosols. The core of the method, that is constituted of the coagulation laws, can be written with a small amount of approximations. However, secondary processes present in planetary atmospheres (sedimentation, electrical charging) need some approximations that alter the quality of the description. Our results show that without electrical charging, the final result of the model described with moments, compared to the reference model described with bins, is better than 5% for the number of aerosols and 20% for the volume in the stratosphere. This is satisfactory considering that, in the frame of climate models, this difference can be easily offseted by other parameters like, for instance, the production rate. When the effect of the electrical charging of the aerosols is considered on the coagulation, the comparison of aerosol numbers between the models are between 10% to a factor three for charging rates between 0 and 45 $e^- \mu m^{-1}$. The mass is generally well represented with a difference less than 20%. These discrepancies can be corrected with a slight offset of the electric charging of about 5 $e^- \mu m^{-1}$.

Although some aspects could still be improved, like the transition between the spherical particles and fractal aggregates, we now consider that the development of the microphysics scheme in terms of moments, for the aerosols microphysics (this work) and the cloud microphysics (Burgalat et al., 2014) is completed. The next step is to use this new model in the TGCM and make comparisons with the version of the TGCM where the old scheme is used. The two-moments scheme model presented in this work is currently 22 times faster than the reference sectional model (100 bins) used for the comparisons. In the frame of a GCM, where the number of bins is reduced (generally around 40 bins to keep a good accuracy), this would lead to a factor 8.8. Since the number of quantities to deal with for the microphysics is highly reduced (4 moments only), execution times are expected to be reduced in all the parts where particles are treated in the GCM, and especially in the dynamical processes.

Acknowledgements

We thank the Agence Nationale de la Recherche (ANR Project "APOSTIC" no 11BS56002, France) for their financial support.

Appendix A. Final set of equations for the Brownian coagulation

From Eq. (21), we have derived the time evolution equation of M_0^S. This is done by applying the harmonic mean method on each terms of the equation. Using Eqs. (22)–(24) in the same way gives the final equations for each selected moment order for the two mode of particles:

$$\frac{dM_0^S}{dt} = \frac{(p_0^{CO}-2)\gamma_0^{CO,SS} \times (p_0^{FM}-2)\gamma_0^{FM,SS}}{(p_0^{CO}-2)\gamma_0^{CO,SS} + (p_0^{FM}-2)\gamma_0^{FM,SS}} \times M_0^{S^2} - \frac{\gamma_0^{CO,SF} \times \gamma_0^{FM,SF}}{\gamma_0^{CO,SF} + \gamma_0^{FM,SF}} \times M_0^S M_0^F$$

$$\frac{dM_3^S}{dt} = \frac{(p_3^{CO}-1)\gamma_3^{CO,SS} \times (p_3^{FM}-1)\gamma_3^{FM,SS}}{(p_3^{CO}-1)\gamma_3^{CO,SS} + (p_3^{FM}-1)\gamma_3^{FM,SS}} \times M_3^{S^2} - \frac{\gamma_3^{CO,SF} \times \gamma_3^{FM,SF}}{\gamma_3^{CO,SF} + \gamma_3^{FM,SF}} \times M_3^S M_3^F$$

$$\frac{dM_0^F}{dt} = \frac{(1-p_0^{CO})\gamma_0^{CO,SS} \times (1-p_0^{FM})\gamma_0^{FM,SS}}{(1-p_0^{CO})\gamma_0^{CO,SS} + (1-p_0^{FM})\gamma_0^{FM,SS}} \times M_0^{S^2} - \frac{\gamma_0^{CO,FF} \times \gamma_0^{FM,FF}}{\gamma_0^{CO,FF} + \gamma_0^{FM,FF}} \times M_0^{F^2}$$

$$\frac{dM_3^F}{dt} = \frac{(1-p_3^{CO})\gamma_3^{CO,SS} \times (p_3^{FM}-1)\gamma_3^{FM,SS}}{(1-p_3^{CO})\gamma_3^{CO,SS} + (p_3^{FM}-1)\gamma_3^{FM,SS}} \times M_3^{S^2} + \frac{\gamma_3^{CO,SF} \times \gamma_3^{FM,SF}}{\gamma_3^{CO,SF} + \gamma_3^{FM,SF}} \times M_3^S M_3^F$$

where

$$
\begin{aligned}
\gamma_0^{CO,SS} &= \left[1 + \alpha_S(1)\alpha_S(-1) + \frac{C}{r_{C_S}}(\alpha_S(-1) + \alpha_S(1)\alpha_S(-2))\right] \times K_{CO}\gamma_0^{CO,SF} \\
&= \left[2 + \alpha_S(1)\, r_{C_S}\, \alpha_F\left(-\frac{3}{D_f}\right) r_{C_F}^{-3/D_f}\, E^{-1} + \alpha_S(1)\, r_{C_S}^{-1}\, \alpha_F\left(\frac{3}{D_f}\right) r_{C_F}^{3/D_f}\, E\right. \\
&\quad \left. + C \times \left(\alpha_S(-1)\, r_{C_S}^{-1} + \alpha_F\left(-\frac{3}{D_f}\right) r_{C_F}^{-3/D_f}\, E^{-1} + \alpha_S(-2)\, r_{C_S}^{-2}\, \alpha_F\left(\frac{3}{D_f}\right) r_{C_F}^{3/D_f}\, E + \alpha_S(1)\, r_{C_S}\alpha_F\left(-\frac{6}{D_f}\right) r_{C_F}^{-6/D_f}\, E^{-2}\right) E^{-2}\right] \times K_{CO}\gamma_0^{CO,FF} \\
&= \left[1 + \alpha_F\left(\frac{3}{D_f}\right)\alpha_F\left(-\frac{3}{D_f}\right) + C\, r_{C_F}^{-3/D_f}\, E^{-1}\left(\alpha_F\left(-\frac{3}{D_f}\right) + \alpha_F\left(\frac{3}{D_f}\right)\alpha_F\left(-\frac{6}{D_f}\right)\right)\right] \times K_{CO}\gamma_0^{FM,SS} \\
&= \left[\alpha_S\left(\frac{1}{2}\right) + 2\,\alpha_S(1)\alpha_S\left(-\frac{1}{2}\right) + \alpha_S(2)\alpha_S\left(-\frac{3}{2}\right)\right] \times\, r_{C_S}^{1/2}\, b_0^{(1)}\, K_{FM}\gamma_0^{FM,SF} \\
&= \left[\alpha_S(1/2) r_{C_S}^{1/2} + 2\,E\,\alpha_S\left(-\frac{1}{2}\right) r_{C_S}^{-1/2}\, \alpha_F\left(\frac{3}{D_f}\right) r_{C_F}^{3/D_f} + E^2\,\alpha_S\left(-\frac{3}{2}\right) r_{C_S}^{-3/2}\, \alpha_F\left(\frac{6}{D_f}\right) r_{C_F}^{6/D_f} + \alpha_S(2) r_{C_S}^{2}\, \alpha_F\left(-\frac{3}{2}\right) r_{C_F}^{-3/2}\right. \\
&\quad \left. + 2\,E\,\alpha_S(1) r_{C_S}\, \alpha_F\left(\frac{6 - 3D_f}{2D_f}\right) r_{C_F}^{\frac{6-3D_f}{2D_f}} + E^2\,\alpha_F\left(\frac{12 - 3D_f}{2D_f}\right) r_{C_F}^{\frac{12-3D_f}{2D_f}}\right] \times b_0^{(4)}\, K_{FM}\gamma_0^{FM,FF} \\
&= \left[\alpha_F\left(\frac{12 - 3D_f}{2D_f}\right) + 2\alpha_F\left(\frac{3}{D_f}\right)\alpha_F\left(\frac{6 - 3D_f}{2D_f}\right) + \alpha_F\left(-\frac{3}{2}\right)\alpha_F\left(\frac{6}{D_f}\right)\right] \times r_{C_F}^{\frac{12-3D_f}{2D_f}}\, E^2 \times b_0^{(3)}\, K_{FM}
\end{aligned}
$$

$$
\begin{aligned}
\gamma_3^{CO,SS} &= \left[2\alpha_S(3) + \alpha_S(2)\alpha_S(1) + \alpha_S(4)\alpha_S(-1) + \frac{C}{r_{C_S}}(\alpha_S(1)^2 + \alpha_S(4)\alpha_S(-2) + \alpha_S(3)\alpha_S(-1) + \alpha_S(2))\right] \times \frac{K_{CO}}{\alpha_S(3)^2\, r_{C_S}^3}\gamma_3^{CO,SF} \\
&= \left[2\alpha_S(3)\, r_{C_S}^3 + \alpha_S(4)\, r_{C_S}^4\, \alpha_F\left(-\frac{3}{D_f}\right) r_{C_F}^{-3/D_f}\, E^{-1} + \alpha_S(2)\, r_{C_S}^2\, \alpha_F\left(\frac{3}{D_f}\right) r_{C_F}^{3/D_f}\, E\right. \\
&\quad \left. + C \times \left(\alpha_S(2)\, r_{C_S}^2 + \alpha_S(3) r_{C_S}^3 \alpha_F\left(-\frac{3}{D_f}\right) r_{C_F}^{-3/D_f}\, E^{-1} + \alpha_S(1)\, r_{C_S}\, \alpha_F\left(\frac{3}{D_f}\right) r_{C_F}^{3/D_f}\, E + \alpha_S(4)\, r_{C_S}^4\, \alpha_F\left(-\frac{6}{D_f}\right) r_{C_F}^{-6/D_f}\, E^{-2}\right)\right] \times \\
&\quad \frac{K_{CO}}{\alpha_S(3)\alpha_F(3) r_{C_S}^3 r_{C_F}^3}\gamma_3^{FM,SS} = \left[\alpha_S\left(\frac{7}{2}\right) + 2\,\alpha_S(1)\alpha_S\left(\frac{5}{2}\right) + \alpha_S(2)\alpha_S\left(\frac{3}{2}\right) + \alpha_S(5)\alpha_S\left(-\frac{3}{2}\right) + 2\,\alpha_S(4)\alpha_S\left(-\frac{1}{2}\right) + \alpha_S(3)\alpha_S\left(\frac{1}{2}\right)\right] \\
&\quad \times \frac{b_3^{(1)}\, K_{FM}}{r_{C_S}^{5/2}\alpha_S(3)^2}\gamma_3^{FM,SF} = \left[\alpha_S\left(\frac{7}{2}\right) r_{C_S}^{7/2} + 2\,E\,\alpha_S\left(\frac{5}{2}\right) r_{C_S}^{5/2}\, \alpha_F\left(\frac{3}{D_f}\right) r_{C_F}^{3/D_f} + E^2\,\alpha_S\left(\frac{3}{2}\right) r_{C_S}^{3/2}\, \alpha_F\left(\frac{6}{D_f}\right) r_{C_F}^{6/D_f} + \alpha_S(5) r_{C_S}^5\, \alpha_F\left(-\frac{3}{2}\right) r_{C_F}^{-3/2}\right. \\
&\quad \left. + 2\,E\,\alpha_S(4) r_{C_S}^4\, \alpha_F\left(\frac{6 - 3D_f}{2D_f}\right) r_{C_F}^{\frac{6-3D_f}{2D_f}} + E^2\,\alpha_S(3) r_{C_S}^3\, \alpha_F\left(\frac{12 - 3D_f}{2D_f}\right) r_{C_F}^{\frac{12-3D_f}{2D_f}}\right] \times \frac{b_3^{(4)}\, K_{FM}}{\alpha_S(3)\alpha_F(3) r_{C_S}^3\, r_{C_F}^3}
\end{aligned}
$$

References

Barth, E. L., & Toon, O. B. (2004). Properties of methane clouds on Titan: Results from microphysical modeling. *Geophysical Research Letters*, 31, 17.

Borucki, W., Whitten, R., Bakes, E., Barth, E., & Tripathi, S. (2006). Predictions of the electrical conductivity and charging of the aerosols in Titan's atmosphere. *Icarus*, 181, 527–544.

Borucki, W. J., Levin, Z., Whitten, R. C., Keesee, R. G., Capone, L. A., Summers, A. L. et al. (1987). Predictions of the electrical conductivity and charging of the aerosols in Titan's atmosphere. *Icarus*, 72, 604–622.

Burgalat, J., Rannou, P., Cours, T., & Rivière, E. (2014). Modeling cloud microphysics using a two-moments hybrid bulk/bin scheme for use in Titans climate models: Application to the annual and diurnal cycles. *Icarus*, 231, 310–322.

Cabane, M., Chassefiere, E., & Israel, G. (1992). Formation and growth of photochemical aerosols in Titan's atmosphere. *Icarus*, 96, 176–189.

Cabane, M., Rannou, P., Chassefiere, E., & Israel, G. (1993). Fractal aggregates in Titan's atmosphere. *Planetary and Space Science*, 41, 257–267.

Cours, T., Burgalat, J., Rannou, P., Rodriguez, S., Brahic, A., & West, R. A. (2011). Dual origin of aerosols in Titan's detached haze layer. *Astrophysical Journal Letters*, 741, L32.

Fiadeiro, M. E., & Veronis, G. (1977). On weighted-mean schemes for the finite-difference approximation to the advection–diffusion equation. *Tellus*, 29, 512–522.

Friedlander, S. K. (2000). *Smoke, dust, and haze: Fundamentals of aerosol dynamics* (2nd ed.) USA: Oxford University Press.

Fuchs, N. A. (1964). *The mechanics of aerosols*. Pergamon Press.

Kahre, M. A., Murphy, J. R., Haberle, R. M., Montmessin, F., & Schaeffer, J. (2005). Simulating the Martian dust cycle with a finite surface dust reservoir. *Geophysical Research Letters*, 32, 20204.

Larson, E. J., Toon, O. B., & Friedson, A. J. (2014). Simulating Titans aerosols in a three dimensional general circulation model. *Icarus*, 243, 400–419.

Larson, E. J., Toon, O. B., West, R. A., & Friedson, A. J. (2015). Microphysical modeling of titans detached haze layer in a 3d gcm. *Icarus*, 254, 122–134.

Lebonnois, S., Burgalat, J., Rannou, P., & Charnay, B. (2012). Titan global climate model: A new 3-dimensional version of the IPSL Titan GCM. *Icarus*, 218, 707–722.
Lee, K. W., Chen, J., & Gieseke, J. A. (1984). Log-normally preserving size distribution for Brownian coagulation in the free-molecule regime. *Aerosol Science and Technology*, 3, 53–62.
Meyers, M. P., Walko, R. L., Harrington, J. Y., & Cotton, W. R. (1997). New RAMS cloud microphysics parameterization. Part II: The two-moment scheme. *Atmospheric Research*, 45, 3–39.
Park, S. H., Lee, K. W., Otto, E., & Fissan, H. (1999). The log-normal size distribution theory of Brownian aerosol coagulation for the entire particle size range: Part I—Analytical solution using the harmonic mean coagulation kernel. *Journal of Aerosol Science*, 30, 3–16.
Pratsinis, S. E. (1988). Simultaneous nucleation, condensation, and coagulation in aerosol reactors. *Journal of Colloid and Interface Science*, 124, 416–427.
Pruppacher, H. R., & Klett, J. D. (2000). *Microphysics of clouds and precipitation*. D. Rediel Publishing Company.
Rannou, P., Hourdin, F., & McKay, C. P. (2002). A wind origin for Titan's haze structure. *Nature*, 418, 853–856.
Rannou, P., Hourdin, F., McKay, C. P., & Luz, D. (2004). A coupled dynamics-microphysics model of Titan's atmosphere. *Icarus*, 170, 443–462.
Rannou, P., Montmessin, F., Hourdin, F., & Lebonnois, S. (2006). The latitudinal distribution of clouds on Titan. *Science*, 311, 201–205.
Tomasko, M. G., Doose, L., Engel, S., Dafoe, L. E., West, R., Lemmon, M. et al. (2008). A model of Titan's aerosols based on measurements made inside the atmosphere. *Planetary and Space Science*, 56, 669–707.
Toon, O. B., McKay, C. P., Courtin, R., & Ackerman, T. P. (1988). Methane rain on Titan. *Icarus*, 75, 255–284.
Toon, O. B., Turco, R. P., Westphal, D., Malone, R., & Liu, M. S. (1988). A multidimensional model for aerosols—Description of computational analogs. *Journal of the Atmospheric Sciences*, 45, 2123–2143.
West, R. A., & Smith, P. H. (1991). Evidence for aggregate particles in the atmospheres of Titan and Jupiter. *Icarus*, 90, 330–333.
Whitby, E. R., & McMurry, P. H. (1997). Modal aerosol dynamics modeling. *Aerosol Science and Technology*, 27, 673–688.